LEVEL 29

Chapter 27: Interview Question Bank

System Design Master Roadmap — 9 Years of Experience Edition

Compiled August 2026

Chapter 27: Interview Question Bank (270+ Questions)

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Delivered as reference tables rather than long-form write-ups for each — this keeps the bank genuinely usable as a practice-planning tool. Pick a row, set a timer matching the "Duration" column, and run the 10-step framework solo before checking whether your answer covers what's in "Interviewer Expects." Questions marked with a chapter reference have a fully worked answer there — the rest are deliberately unworked, for genuine independent practice.

27.1 Beginner (50)

#	Question	Concepts Tested	Duration	Interviewer Expects
1	Design a URL shortener *(Ch.16)*	ID generation, cache-aside, read-heavy scaling	30 min	Base62 encoding reasoning, cache justification
2	Design Pastebin *(Ch.16)*	Metadata/blob split, expiry	30 min	Object storage vs. inline storage threshold reasoning
3	Design a rate limiter *(Ch. 16)*	Token bucket, atomic Redis ops	30 min	Algorithm choice + fail-open/closed trade-off
4	Design a file upload service *(Ch.16)*	Pre-signed URLs, async processing	30 min	Direct-to-storage upload pattern
5	Design a single-channel notification service	Queueing, retries	30 min	Async decoupling from trigger
6	Design a log aggregation system *(Ch.16)*	Buffering, indexing pipeline	35 min	App never blocks on logging
7	Design a parking lot system (LLD)	OOP design, state management	35 min	Class structure, spot-assignment logic
8	Design an elevator system (LLD)	State machine, scheduling algorithm	35 min	Request-handling algorithm clarity
9	Design a vending machine (LLD)	State machine	30 min	Clean state transitions
10		Entity relationships, concurrency on checkout	35 min	Preventing double-checkout of one copy

#	Question	Concepts Tested	Duration	Interviewer Expects
	Design a library management system (LLD)			
11	Design an ATM (LLD)	State machine, concurrency	35 min	Atomic balance operations
12	Design tic-tac-toe (LLD)	OOP, win-condition logic	25 min	Clean extensibility (e.g., to N x N board)
13	Design an in-memory key-value store	Hash map, eviction	25 min	Data structure choice reasoning
14	Design an LRU cache	Data structure (hashmap + doubly linked list)	25 min	O(1) get/put reasoning
15	Design a to-do list backend	Basic CRUD, data modeling	20 min	Clean REST resource design
16	Design a polling/voting system (small scale)	Data modeling, double-vote prevention	30 min	Uniqueness constraint reasoning
17	Design a basic blogging platform	CRUD, pagination	30 min	Cursor pagination justification
18	Design a weather app backend	Caching third-party API responses	25 min	TTL-based caching of upstream data
19	Design a simple shopping cart (no checkout)	Session/state modeling	25 min	Cart persistence approach
20	Design 1:1 chat with no scale requirement	Basic messaging, WebSocket intro	30 min	Simple send/receive flow
21	Design a health-check endpoint strategy	Liveness vs. readiness	15 min	Correct distinction (Ch. 8.4)
22	Design a basic login/ signup system	Password hashing, sessions	30 min	bcrypt/argon2 reasoning
23	Design a simple leaderboard	Sorted data structure	25 min	Redis sorted set reasoning
24	Design a coupon/discount code system	Uniqueness, expiry, usage limits	30 min	Atomic redemption-count check
25	Design an event RSVP system	Capacity limits, concurrency	30 min	Atomic capacity-check-and-reserve
26	Design a basic image resizing service	Async processing	25 min	Queue-triggered worker pattern
27	Design a simple form/ survey builder backend	Dynamic schema modeling	30 min	Flexible schema decision (JSON column vs. doc DB)
28			25 min	Atomic conditional update

#	Question	Concepts Tested	Duration	Interviewer Expects
	Design a single-store inventory tracker	Basic CRUD, atomic decrement		
29	Design a basic appointment scheduler	Time-slot conflict prevention	30 min	Overlap-prevention logic
30	Design a basic blog comment system	Nested data, pagination	25 min	Threaded-comment data modeling
31	Design a background job queue	Async processing, retries	30 min	Queue + worker + DLQ pattern
32	Design a website uptime monitor	Polling, alerting	30 min	Scheduled health checks + alert thresholds
33	Design a feature-flag toggle service (single-region)	Config propagation	25 min	Fast read path, cache-backed
34	Design a basic audit log system	Append-only logging	25 min	Immutability reasoning
35	Design pastebin with auto-expiry	TTL-based cleanup	20 min	Lazy vs. scheduled cleanup trade-off
36	Design a session management system	Stateless vs. stateful tokens	30 min	JWT vs. session trade-off (Ch.1.5)
37	Design a contact-form/ lead-capture backend	Simple ingestion, spam prevention	20 min	Rate limiting on submission
38	Design a content tagging system	Many-to-many modeling	25 min	Join table / tag index design
39	Design a bookmarks/ favorites service	Simple CRUD at user scale	20 min	Efficient per-user listing query
40	Design a basic FAQ search	Simple text search	25 min	When full-text search is/ isn't overkill
41	Design a page-view counter service	High-write counters	25 min	Redis INCR + async flush pattern
42	Design a password reset flow	Security, token expiry	25 min	Time-limited, single-use token
43	Design an email verification flow	Async, token-based confirmation	20 min	Token expiry + resend handling
44	Design an API key management system	Secrets handling	25 min	Hashed storage, scoped permissions
45	Design a config management service	Central config, environment overrides	25 min	Config-as-code vs. dynamic service
46		Tenant isolation	30 min	

#	Question	Concepts Tested	Duration	Interviewer Expects
	Design a multi-tenant feature flag system			Per-tenant override resolution
47	Design a single-instance in-memory rate limiter	Basic algorithm implementation	20 min	Correct token-bucket logic
48	Design a reusable retry library	Backoff, jitter	20 min	Exponential backoff + jitter correctness
49	Design a reusable circuit breaker library	State machine (closed/open/half-open)	25 min	Correct state transitions
50	Design a basic service health dashboard	Aggregating health checks	25 min	Polling vs. push-based health aggregation

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27.2 Intermediate (50)

#	Question	Concepts Tested	Duration	Interviewer Expects
1	Design Twitter/X *(Ch.17)*	Fan-out, celebrity problem	45 min	Hybrid fan-out resolution
2	Design Instagram *(Ch.17)*	Media pipeline + feed	45 min	Upload/CDN + feed fan-out split
3	Design WhatsApp *(Ch.17)*	Connection routing, presence	45 min	Gateway + routing-layer design
4	Design YouTube *(Ch.17)*	Transcoding, CDN	45 min	Bandwidth-first framing
5	Design Dropbox *(Ch.17)*	Chunking, sync, dedup	45 min	Block-level sync reasoning
6	Design Google Drive *(Ch.17)*	Permissions, collaboration	45 min	Permission-check design + OT/CRDT awareness
7	Design an e-commerce system *(Ch.17)*	Catalog vs. checkout split	45 min	Different consistency models per sub-path
8	Design a food delivery system *(Ch.17)*	Geospatial matching, order state	45 min	Location pipeline separate from order data
9	Design hotel booking *(Ch. 17)*	Date-ranged inventory	45 min	No-double-booking atomicity
10	Design ticket booking *(Ch. 17)*	Seat locks, admission control	45 min	TTL-based hold + waiting room
11	Design a basic ride-sharing app	Matching, trip lifecycle	45 min	Matching + state machine
12			40 min	Recency-weighted ranking

#	Question	Concepts Tested	Duration	Interviewer Expects
	Design a job board / job search platform	Search ranking, two-sided matching		
13	Design a video conferencing app (small groups)	Real-time media, signaling	45 min	Signaling server + peer connection basics
14	Design a collaborative code editor	Real-time sync, conflict resolution	45 min	OT/CRDT awareness
15	Design a music streaming service	Catalog + streaming + offline	40 min	CDN-first + licensing awareness
16	Design a podcast platform	Media hosting, RSS-like distribution	35 min	Object storage + CDN reasoning
17	Design a "stories" feature (24h expiry content)	TTL content, view tracking	35 min	Efficient expiry mechanism
18	Design a group chat/ channels feature	Fan-out to many members	40 min	Fan-out-on-write for bounded groups
19	Design a large-scale polling app	High-write counters, race conditions	40 min	Atomic increment + eventual display consistency
20	Design an online auction system	Bidding concurrency, real-time updates	45 min	Atomic highest-bid update + real-time push
21	Design a review/rating aggregation system	Aggregation at scale	35 min	Precomputed aggregate, async update
22	Design a web crawler	Queue-based traversal, dedup	45 min	URL frontier + dedup + politeness/rate limits
23	Design a distributed cron scheduler	Exactly-once-ish scheduling	40 min	Leader election for single-trigger guarantee
24	Design an email service backend	Storage, search, delivery	45 min	Metadata/content split, search indexing
25	Design a calendar system with invites	Conflict detection, notifications	40 min	Overlap detection + async invite delivery
26	Design a survey platform at scale	High-volume writes, analytics	35 min	Write-optimized ingestion, async aggregation
27	Design a content moderation pipeline	Async review, queueing	40 min	Pre/post-publish moderation trade-off
28	Design an A/B testing platform	Bucketing, consistent assignment	40 min	Deterministic hashing for bucket assignment
29	Design a feature-flag system at scale	Propagation latency, caching	35 min	Push vs. pull propagation trade-off
30			40 min	

#	Question	Concepts Tested	Duration	Interviewer Expects
	Design URL shortener + analytics dashboard	Async analytics ingestion		Analytics off the critical redirect path
31	Design a subscription/ recurring billing system	Scheduled charges, retries	40 min	Dunning/retry logic for failed charges
32	Design a multiplayer trivia game	Real-time state sync	40 min	Room-based state + WebSocket fan-out
33	Design a live event Q&A app (Slido-like)	Real-time fan-out, moderation	35 min	Broadcast to session attendees
34	Design a Google-Docs-lite editor	Concurrent editing	45 min	OT/CRDT reasoning
35	Design a classifieds marketplace	Listings, search, messaging	40 min	Search + messaging integration
36	Design a car rental system	Inventory across locations, date ranges	40 min	Date-ranged inventory (echoes hotel booking)
37	Design a coworking space booking system	Resource + time-slot booking	35 min	Same atomicity pattern as hotel booking
38	Design a fitness tracking app backend	High-volume sensor-like data ingestion	35 min	Time-series storage reasoning
39	Design a recipe-sharing platform	Search, media, ratings	35 min	Standard CRUD + search integration
40	Design a crowdfunding platform	Payment + goal tracking	40 min	Atomic pledge tracking, payment integration
41	Design a Reddit-like forum	Voting, ranking, nested comments	45 min	Hot/top ranking algorithm
42	Design a Stack-Overflow-like Q&A platform	Search, reputation, voting	40 min	Search + ranking integration
43	Design video call recording/ storage	Large media ingestion + storage	35 min	Async post-call processing pipeline
44	Design a digital signage/ content distribution system	Scheduled content push	30 min	CDN-based content delivery to devices
45	Design a loyalty points/ rewards system	Points ledger, redemption	40 min	Ledger-style point tracking (echoes Ch.20)
46	Design a multi-vendor inventory sync system	Cross-system consistency	40 min	Eventual consistency + reconciliation
47	Design a shipment tracking system	Status updates, notifications	35 min	State machine + async notification
48	Design a clinic appointment booking system	Time-slot conflict, reminders	35 min	Conflict prevention + scheduled reminders

#	Question	Concepts Tested	Duration	Interviewer Expects
49	Design a city-wide parking-space finder	Real-time availability, geospatial	40 min	Geospatial index (echoes Ch.18)
50	Design a lost-and-found platform	Search/matching	30 min	Simple matching/search design

27.3 Advanced (50)

#	Question	Concepts Tested	Duration	Interviewer Expects
1	Design Uber *(Ch.18)*	Matching, surge, trip state machine	45–60 min	Geospatial matching + surge pricing design
2	Design Careem *(Ch.18)*	Super-app, shared platform services	45–60 min	Vertical vs. platform service boundary
3	Design Amazon *(Ch.18)*	Marketplace, fulfillment, Buy Box	60 min	Multi-tenant + fulfillment routing
4	Design Netflix *(Ch.18)*	Personalization, licensing, CDN	60 min	Precomputed ranking + regional licensing model
5	Design a Payment Gateway *(Ch.18, Ch.20)*	Idempotency, state machine, reconciliation	60 min	Unknown-outcome handling
6	Design a Digital Wallet *(Ch. 18, Ch.20)*	Double-entry ledger	45–60 min	Derived balance reasoning
7	Design a Ride Matching Engine *(Ch.18)*	Geospatial index, bipartite matching	45 min	Filter-then-rank pattern
8	Design Food Delivery at scale *(Ch.18)*	Order batching, fleet optimization	45 min	Batching scoring function
9	Design a web-scale search engine *(Ch.22)*	Inverted index, ranking, crawling	60 min	Indexing pipeline + ranking depth
10	Design a Recommendation System *(Ch.18)*	Candidate generation + ranking	45–60 min	Two-stage architecture
11	Design an Ad Serving System *(Ch.18)*	Auction, budget pacing	45–60 min	Latency budget + pacing design
12	Design a Distributed Notification System *(Ch.18)*	Multi-region, priority tiers	45 min	Separate lanes for critical vs. bulk
13	Design Real-time Chat at scale (Slack/Discord) *(Ch.18)*	Large-channel fan-out	45–60 min	Sparse connection-state reasoning
14	Design Live Video Streaming (Twitch) *(Ch.18)*	Live ingest, segmented delivery	45–60 min	Latency/quality/scale trade-off

#	Question	Concepts Tested	Duration	Interviewer Expects
15	Design a global CDN	Edge caching, origin shielding	45 min	Multi-tier caching to protect origin
16	Design a distributed cache (Redis Cluster from scratch)	Consistent hashing, replication	45–60 min	Ring-based partitioning + failover
17	Design a multi-region distributed rate limiter	Coordination trade-offs	45 min	Local approximation vs. global precision
18	Design a distributed lock service	Consensus, lease/TTL	45 min	Correctness under node failure
19	Design a distributed ID generator (Snowflake-style)	Coordination-free uniqueness	30 min	Reserved-range/ timestamp-based design
20	Design a distributed job scheduler at scale	Exactly-once execution	45 min	Leader election + idempotent execution
21	Design a global load balancer/traffic manager	GeoDNS, health-based routing	40 min	Multi-layer routing decision
22	Design a stock exchange/ trading system	Order matching, extreme low latency	60 min	Order book + matching engine design
23	Design a fraud detection pipeline	Real-time + batch scoring	45 min	Two-stage inline/async pattern
24	Design a global search autocomplete system	Prefix indexing at scale	40 min	Edge n-gram / trie reasoning
25	Design a large-scale log analytics platform	Ingestion, indexing, retention	45 min	Buffer-then-index pipeline
26	Design a real-time analytics/ metrics platform	Streaming aggregation	45 min	Windowed aggregation design
27	Design an ML feature store	Training-serving consistency	45 min	Training-serving skew awareness
28	Design a billion-user push notification system	Extreme fan-out	45–60 min	Priority-tiered, sharded delivery
29	Design a multi-region active-active database	Conflict resolution, consensus	60 min	CRDT/last-write-wins/ quorum trade-offs
30	Design a global rate-limited API gateway	Edge enforcement	40 min	Local + global limit coordination
31	Design Google Maps / a routing service	Graph routing, live traffic	60 min	Road-graph + live-traffic overlay
32	Design a distributed task queue at scale	Backpressure, worker scaling	45 min	Queue-depth-driven autoscaling
33		Transcoding + delivery	45 min	

#	Question	Concepts Tested	Duration	Interviewer Expects
	Design a large-scale image/ video CDN pipeline			Async processing + tiered caching
34	Design a global experimentation platform	Consistent bucketing at scale	45 min	Deterministic hashing, statistical validity awareness
35	Design a real-time payment fraud/anomaly system	Streaming ML scoring	45–60 min	Inline rules + async model split
36	Design a distributed tracing system (Jaeger-like)	Span collection, sampling	45 min	Sampling strategy at scale
37	Design a metrics monitoring system (Prometheus-like)	Time-series storage, scraping	45 min	Pull vs. push model trade-off
38	Design a global CDN cache-invalidation system	Propagation consistency	40 min	Versioned URLs vs. active purge trade-off
39	Design a large-scale transactional email system (SES-like)	Deliverability, reputation	40 min	Queue + provider-rate-limit-aware sending
40	Design a global session store	Multi-region consistency	40 min	Region-local + replicated fallback
41	Design a multi-tenant SaaS platform architecture	Tenant isolation, noisy-neighbor	45–60 min	Isolation strategy (shared vs. siloed data)
42	Design a large-scale ETL/data warehouse pipeline	Batch processing at scale	45 min	Batch vs. stream processing trade-off
43	Design a global URL shortener with abuse detection	Scale + security	40 min	Abuse-pattern detection design
44	Design a large-scale voting/ election system	Integrity, extreme peak load	45–60 min	Correctness-under-load + auditability
45	Design a distributed leader-election service	Consensus (Raft/Paxos)	40 min	Correct failure-handling reasoning
46	Design a global chat presence system	Sparse state at scale	40 min	Lazy/on-demand presence design
47	Design a large-scale price-comparison engine	Cross-source aggregation	40 min	Async ingestion + normalization pipeline
48	Design Amazon's warehouse/ fulfillment routing	Multi-warehouse optimization	45–60 min	Nearest-with-stock routing logic
49	Design a global content ranking pipeline	Precomputation at scale	45 min	Offline ranking + fast-serving split
50			60 min	

#	Question	Concepts Tested	Duration	Interviewer Expects
	Design multi-region disaster recovery for a bank	RPO/RTO at extreme correctness bar		Near-zero-RPO replication design

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27.4 FinTech (30)

Companion reading: Chapter 20, Chapter 18 Problems 21–22.

#	Question	Concepts Tested	Duration	Interviewer Expects
1	Design a payment gateway	Idempotency, state machine	60 min	Full flow incl. unknown-outcome handling
2	Design a digital wallet	Double-entry ledger	45–60 min	Derived-balance reasoning
3	Design a P2P money transfer system	Atomic paired ledger entries	45 min	Single-transaction atomicity
4	Design a double-entry ledger from scratch	Accounting invariants	45 min	Debits-equal-credits invariant explained
5	Design transaction fraud detection	Two-stage scoring	45 min	Inline rules + async ML split
6	Design a reconciliation system	Batch matching, discrepancy handling	45–60 min	Settlement-file matching logic
7	Design a refund processing system	Linked reversing transaction	35 min	Refund as new, linked transaction
8	Design chargeback handling	External-event ingestion, dispute workflow	40 min	Chargeback vs. refund distinction
9	Design a subscription billing system	Scheduled charges, dunning	40 min	Retry/dunning logic for failed renewals
10	Design a BNPL (buy-now-pay-later) system	Installment scheduling, risk	45 min	Scheduled-charge state machine
11	Design a credit scoring system	Data aggregation, model serving	45 min	Feature aggregation + serving latency
12	Design a loan origination system	Multi-step approval workflow	45 min	State machine for approval stages
13	Design invoice generation and tracking	Scheduled generation, status tracking	35 min	Status state machine
14	Design a multi-currency payment system	FX conversion, consistency	45 min	Conversion-rate-locking reasoning

#	Question	Concepts Tested	Duration	Interviewer Expects
15	Design a UPI-like real-time payment system	Low-latency settlement	45–60 min	Real-time rail integration reasoning
16	Design a stock order matching engine	Extreme low latency, order book	60 min	In-memory order book design
17	Design a crypto wallet system	Key management, transaction signing	45 min	Security-first key handling
18	Design a KYC verification pipeline	Async document processing	40 min	Async review + status state machine
19	Design a payment webhook delivery system	Reliable async delivery	40 min	Retry + idempotent receiver design
20	Design a marketplace payout/disbursement system	Batch payouts, ledger accuracy	45 min	Batch payout reconciliation
21	Design a budgeting/ expense tracker backend	Categorization, aggregation	30 min	Efficient per-user aggregation
22	Design an insurance claims processing system	Multi-step workflow, documents	45 min	State machine + async document review
23	Design a tax calculation engine for checkout	Rule-based computation, jurisdictions	40 min	Rule-engine design for jurisdiction variance
24	Design a merchant settlement system	Batch aggregation, ledger	45 min	Aggregation + reconciliation design
25	Design a bill-split app (Splitwise-like)	Multi-party ledger	40 min	Simplified debt-graph/ ledger design
26	Design a rewards/cashback system	Points ledger, expiry rules	35 min	Ledger-style point tracking
27	Design a savings-goal/ fixed-deposit feature	Scheduled accrual	35 min	Interest-accrual scheduling
28	Design an ATM network transaction system	Distributed consistency, offline handling	45 min	Network-partition handling for cash dispensing
29	Design a cross-border remittance system	Multi-hop settlement, compliance	45–60 min	Multi-rail + compliance-check integration
30	Design a financial audit trail system	Immutable logging, queryability	35 min	Append-only + efficient audit query design

27.5 Real-Time Systems (30)

Companion reading: Chapter 21, Chapter 17 Problem 9, Chapter 18 Problems 17, 29, 30.

#	Question	Concepts Tested	Duration	Interviewer Expects
1	Design WhatsApp	Connection routing, ordering	45 min	Gateway + routing-layer pattern
2	Design a live chat support widget	WebSocket, queue routing to agents	35 min	Agent-assignment + real-time transport
3	Design real-time delivery location tracking	UDP-tolerant ingestion, targeted push	40 min	Latest-value-wins + targeted routing
4	Design a live sports score system	One-to-many broadcast	35 min	Compute-once, fan-out-many (SSE)
5	Design multiplayer game state sync	Low-latency state broadcast	45 min	Tick-based state sync reasoning
6	Design a live auction bidding system	Real-time updates + atomicity	40 min	Atomic highest-bid + broadcast
7	Design a real-time collaborative doc editor	OT/CRDT	45 min	Operation-level merge reasoning
8	Design a live ops metrics dashboard	One-to-many broadcast	35 min	Compute-once-fan-out-many pattern
9	Design a presence (online/offline) system	Sparse state, TTL heartbeats	35 min	Lazy pull + TTL-based liveness
10	Design a 1:1 video call system	Signaling, peer connection	40 min	Signaling server design
11	Design group video conferencing	Media routing (SFU/MCU concepts)	45 min	Media-routing architecture awareness
12	Design a live-stream chat overlay	High fan-out chat	40 min	Fan-out-to-thousands pattern
13	Design a real-time stock ticker	High-frequency broadcast	35 min	Throttled/batched broadcast design
14	Design a real-time ride-tracking map	Targeted push, geospatial	40 min	Single-recipient targeted routing
15	Design an in-app notification bell	Targeted push + persistence	30 min	Durable-first, push-second design
16	Design a real-time voting/poll display	Aggregation + broadcast	30 min	Async aggregation, broadcast result
17	Design a typing-indicator feature		25 min	No-persistence-needed reasoning

#	Question	Concepts Tested	Duration	Interviewer Expects
		Ephemeral, high-frequency, low-durability signal		
18	Design a live ticket-sale waiting room	Admission control	40 min	Queue-based admission control
19	Design a real-time fraud alert system	Streaming detection + push	40 min	Streaming pipeline + alert delivery
20	Design a real-time IoT sensor data pipeline	High-throughput ingestion	45 min	Kafka-class ingestion + downstream fan-out
21	Design a real-time game leaderboard	Sorted structure + push updates	35 min	Redis sorted set + targeted broadcast
22	Design real-time flight tracking	High-frequency position updates	40 min	Latest-value-wins ingestion
23	Design real-time inventory sync across warehouses	Eventual consistency at speed	40 min	Event-driven sync + reconciliation
24	Design real-time support-queue routing	Dynamic assignment	35 min	Real-time queue-depth-aware routing
25	Design a real-time bidding (RTB) ad system	Extreme low latency	45–60 min	Sub-100ms auction pipeline
26	Design a real-time GPS fleet management system	High-volume ingestion, geospatial	40 min	Ingestion + geospatial index pattern
27	Design live captioning/ translation	Streaming processing + low latency	40 min	Streaming pipeline with bounded latency
28	Design a real-time collaborative whiteboard	OT/CRDT for shapes	40 min	Operation-based sync for non-text data
29	Design a scalable push notification system	Fan-out + provider rate limits	40 min	Per-channel queue + throttling
30	Design a real-time multiplayer trivia buzzer	Race-condition-free "first buzz wins"	30 min	Atomic first-write-wins logic

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27.6 E-commerce (30)

Companion reading: Chapter 17 Problem 13, Chapter 18 Problem 19.

#	Question	Concepts Tested	Duration	Interviewer Expects
1	Design an e-commerce catalog and search	Read-heavy scaling, search integration	45 min	Cache/CDN + Elasticsearch integration

#	Question	Concepts Tested	Duration	Interviewer Expects
2	Design a shopping cart service	Ephemeral state, fast reads/writes	35 min	Redis/DynamoDB-backed cart reasoning
3	Design a checkout and order system	Strong consistency, Saga	45–60 min	Checkout vs. catalog consistency split
4	Design an inventory management system	Atomic conditional updates	40 min	No-oversell atomicity
5	Design a flash-sale system	Extreme peak load, admission control	45–60 min	Pre-warming + waiting-room pattern
6	Design a product recommendation engine	Two-stage candidate/rank	45 min	Candidate generation + ranking split
7	Design a wishlist/favorites service	Simple CRUD at scale	20 min	Efficient per-user listing
8	Design a review and rating system	Aggregation, abuse prevention	35 min	Precomputed aggregate + fraud reasoning
9	Design seller/marketplace onboarding	Multi-step verification workflow	35 min	State machine for onboarding stages
10	Design a returns/refunds system	Workflow + ledger linkage	40 min	Linked reversing transaction (echoes Ch. 20)
11	Design a shipping/logistics tracking system	Status updates, notifications	40 min	State machine + async notification
12	Design a price-drop alert system	Watch-list matching at scale	35 min	Efficient match-on-price-change design
13	Design a coupon/discount engine	Rule evaluation, atomic redemption	40 min	Atomic usage-limit enforcement
14	Design a loyalty/rewards program	Points ledger	35 min	Ledger-style tracking
15	Design a product Q&A system	CRUD + moderation	30 min	Moderation pipeline integration
16	Design an abandoned-cart recovery system	Scheduled/triggered notifications	30 min	Event-triggered async workflow
17	Design multi-warehouse order fulfillment	Nearest-with-stock routing	45 min	Warehouse-selection logic
18	Design a dynamic pricing engine	Real-time rule evaluation	40 min	Streaming signal ingestion for pricing
19	Design a Buy-Box/offer-ranking system	Multi-factor ranking, caching	40 min	Precomputed, cached ranking

#	Question	Concepts Tested	Duration	Interviewer Expects
20	Design an order-tracking notification system	Status-change-triggered push	30 min	Event-driven notification design
21	Design a subscription box (recurring order) service	Scheduled recurring fulfillment	40 min	Scheduled job + inventory reservation
22	Design a gift card system	Balance tracking, redemption	35 min	Ledger-style balance (echoes Ch.20)
23	Design a size/fit recommendation system	ML-adjacent, feature-based	40 min	Feature-based scoring reasoning
24	Design visual (image) search for a catalog	Embedding-based search	45 min	Vector/embedding search awareness
25	Design a supplier inventory sync system	Cross-system eventual consistency	40 min	Event-driven sync + reconciliation
26	Design fake-order/fraud detection	Two-stage scoring	40 min	Inline + async detection split
27	Design a customer support ticketing system	Queueing, routing, SLAs	35 min	Priority-queue routing
28	Design a live "stock left" counter at scale	High-read, approximate-OK counters	35 min	Cached approximate count reasoning
29	Design a marketplace dispute resolution system	Multi-party workflow	40 min	State machine across buyer/seller/platform
30	Design a bulk product-catalog import pipeline	Batch processing, validation	40 min	Async batch validation + partial-failure handling

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27.7 Distributed Systems (30)

Companion reading: Chapter 6, Chapter 13.

#	Question	Concepts Tested	Duration	Interviewer Expects
1	Design a distributed lock service	Consensus, lease/TTL	45 min	Correctness under node failure
2	Design a leader election service	Raft/Paxos-based consensus	40 min	Correct failure-handling
3	Design a distributed consensus system	Raft mechanics	45–60 min	Understanding of quorum-based agreement
4	Design a distributed unique ID generator	Coordination-free uniqueness	30 min	Reserved-range/Snowflake-style design

#	Question	Concepts Tested	Duration	Interviewer Expects
5	Design a distributed rate limiter	Local vs. global coordination	40 min	Precision/latency trade-off
6	Design a distributed cache with consistent hashing	Ring-based partitioning	45 min	Minimal-remap reasoning
7	Design a distributed transaction coordinator (Saga)	Compensating transactions	45 min	Orchestration vs. choreography trade-off
8	Design an outbox-pattern event publisher	Atomic DB write + publish	35 min	Single-transaction outbox reasoning
9	Design a change-data-capture pipeline	Log-tailing, downstream sync	40 min	CDC vs. dual-write reasoning
10	Design a distributed job scheduler (exactly-once)	Leader election + idempotency	45 min	Idempotent execution under retries
11	Design a distributed configuration service	Propagation, consistency	35 min	Push/pull propagation trade-off
12	Design a distributed tracing system	Span collection, sampling	45 min	Sampling strategy at scale
13	Design a gossip-protocol membership service	Eventually-consistent membership	40 min	Gossip convergence reasoning
14	Design a distributed queue with exactly-once processing	Delivery semantics	40 min	At-least-once + idempotency = effective exactly-once
15	Design multi-region data replication	Conflict resolution	45–60 min	CRDT/LWW/quorum trade-offs
16	Design a CRDT-based distributed counter	Conflict-free merge	35 min	Merge-function correctness
17	Design a simplified distributed file system	Chunking, replication, metadata	60 min	Metadata/chunk-server split (HDFS-style)
18	Design a quorum-based distributed key-value store	N/W/R tuning	40 min	Correct quorum math
19	Design a distributed token-bucket rate-limiting service	Shared state coordination	35 min	Redis-backed atomic bucket design
20	Design a service mesh control plane	Config distribution to sidecars	45 min	Control plane/data plane separation
21	Design distributed circuit-breaker coordination	Shared failure-state signaling	35 min	Per-instance vs. shared breaker state trade-off

#	Question	Concepts Tested	Duration	Interviewer Expects
22	Design a distributed deduplication service	Idempotency at scale	35 min	Efficient seen-before check (e.g., bloom filter)
23	Design a distributed session store	Multi-region consistency	35 min	Region-local + replication trade-off
24	Design distributed feature-flag propagation	Fast, consistent rollout	35 min	Propagation-latency vs. consistency trade-off
25	Design a guaranteed-delivery distributed logging pipeline	Durability under failure	40 min	Buffer-before-index reasoning
26	Design a distributed batch-processing framework	Partitioned parallel processing	45 min	Map-reduce-style partitioning
27	Design leader-follower DB replication from scratch	Replication lag, failover	45 min	Failover mechanics + lag handling
28	Design a distributed event bus with schema evolution	Compatibility management	40 min	Backward-compatible schema versioning
29	Design a multi-region rate-limiting proxy	Global vs. local enforcement	40 min	Budget-allocation-per-region pattern
30	Design a distributed clock synchronization approach	Clock skew, logical clocks	35 min	Logical clocks / TrueTime-style awareness

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Next → Chapter 28: Mock Interview Program — 20 full mock interviews (interviewer version + candidate version) spanning beginner to Tier-1 difficulty.